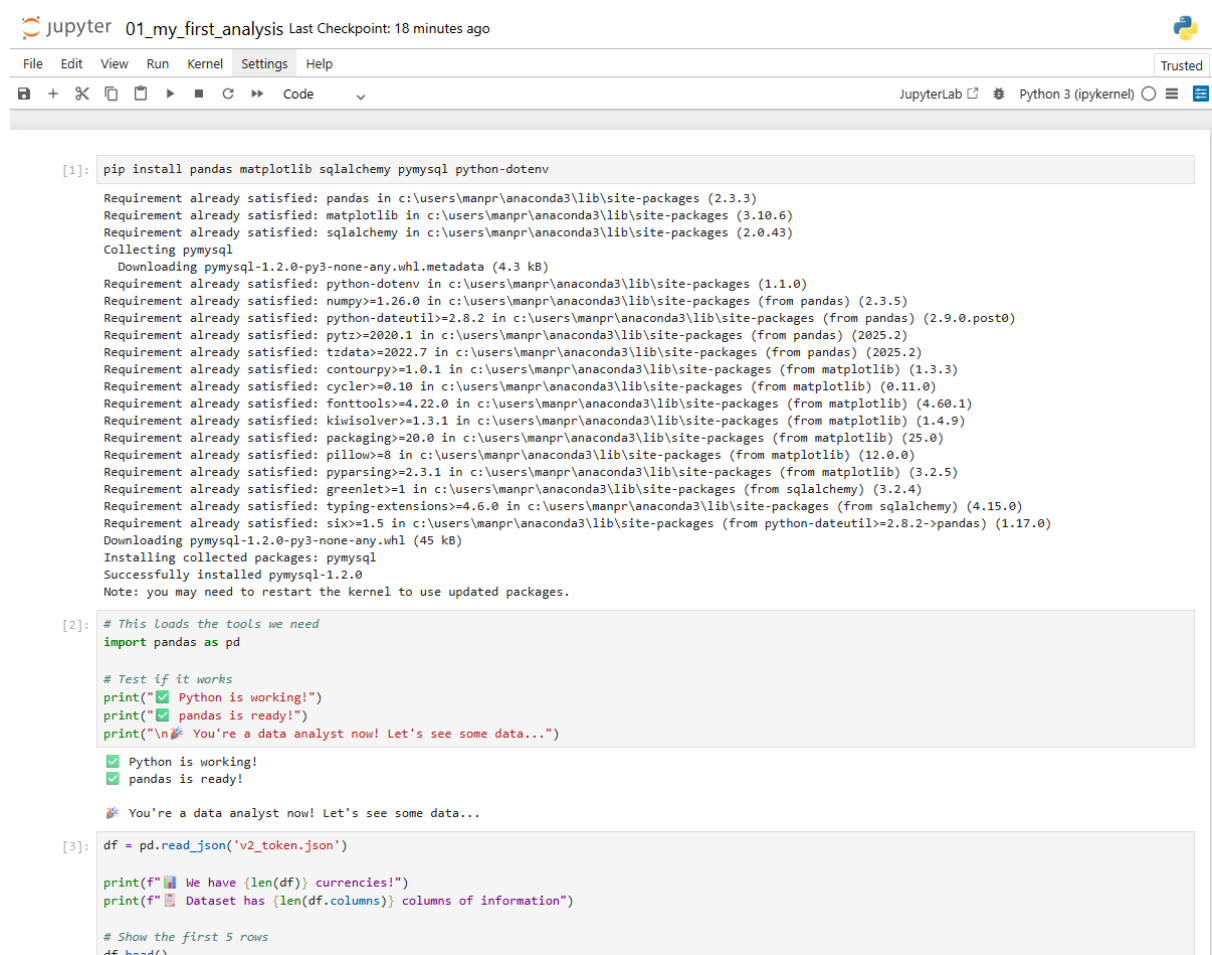


1. Your Jupyter notebook with all cells run



```
[1]: pip install pandas matplotlib sqlalchemy pymysql python-dotenv

Requirement already satisfied: pandas in c:\users\manpr\anaconda3\lib\site-packages (2.3.3)
Requirement already satisfied: matplotlib in c:\users\manpr\anaconda3\lib\site-packages (3.10.6)
Requirement already satisfied: sqlalchemy in c:\users\manpr\anaconda3\lib\site-packages (2.0.43)
Collecting pymysql
  Downloading pymysql-1.2.0-py3-none-any.whl.metadata (4.3 kB)
Requirement already satisfied: python-dotenv in c:\users\manpr\anaconda3\lib\site-packages (1.1.0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\manpr\anaconda3\lib\site-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\manpr\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\manpr\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\manpr\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\manpr\anaconda3\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\manpr\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\manpr\anaconda3\lib\site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\manpr\anaconda3\lib\site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\users\manpr\anaconda3\lib\site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\users\manpr\anaconda3\lib\site-packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\manpr\anaconda3\lib\site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: greenlet>=1 in c:\users\manpr\anaconda3\lib\site-packages (from sqlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\manpr\anaconda3\lib\site-packages (from sqlalchemy) (4.15.0)
Requirement already satisfied: six>=1.5 in c:\users\manpr\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading pymysql-1.2.0-py3-none-any.whl (45 kB)
Installing collected packages: pymysql
Successfully installed pymysql-1.2.0
Note: you may need to restart the kernel to use updated packages.
```

```
[2]: # This loads the tools we need
import pandas as pd

# Test if it works
print("✅ Python is working!")
print("✅ pandas is ready!")
print("\n🎉 You're a data analyst now! Let's see some data...")

✅ Python is working!
✅ pandas is ready!

🎉 You're a data analyst now! Let's see some data...
```

```
[3]: df = pd.read_json('v2_token.json')

print(f"📄 We have {len(df)} currencies!")
print(f"📊 Dataset has {len(df.columns)} columns of information")

# Show the first 5 rows
df.head()
```

```
[3]: df = pd.read_json('v2_token.json')

print(f"📄 We have {len(df)} currencies!")
print(f"📄 Dataset has {len(df.columns)} columns of information")

# Show the first 5 rows
df.head()
```

📄 We have 78 currencies!
📄 Dataset has 41 columns of information

```
[3]:
```

	_id	symbol	base_currency	base_currency_desc	base_currency_logoid	circulating_supply	crypto_blockchain_ecosystems_tr	crypto_comm
--	-----	--------	---------------	--------------------	----------------------	--------------------	---------------------------------	-------------

0	67b761fd00bd4e4006afb817	DJEDUSD	DJED	Djed	crypto/XTVCDJED	3293404.07		[Stablecc
---	--------------------------	---------	------	------	-----------------	------------	--	-----------

1	68b029941f7ecf768350484f	USDTUSD	USDT	Tether USDT	crypto/XTVCUSDT	186150733914.479614	[Ethereum ecosystem, Polygon ecosystem, Tron e...	[Stablecc st
---	--------------------------	---------	------	-------------	-----------------	---------------------	---	--------------

2	67b761f800bd4e4006afb80f0	USDCUSD	USDC	USDC	crypto/XTVCUSDC	78603600504.580612	[Ethereum ecosystem, Solana ecosystem, Polygon...	[Stablecc sta
---	---------------------------	---------	------	------	-----------------	--------------------	---	---------------

3	67b761f900bd4e4006afb8753	USDS2USD	USDS	USDS	crypto/XTVCUSDS2	9912848232.125813		[Stablecc stat
---	---------------------------	----------	------	------	------------------	-------------------	--	----------------

4	67b761f800bd4e4006afb8178	USDEEUSD	USDE	Ethena USDe	crypto/XTVCUSDEE	6559535736.529217	[EigenLayer ecosystem]	[Stablecc
---	---------------------------	----------	------	-------------	------------------	-------------------	------------------------	-----------

5 rows × 41 columns

Jupyter 01_my_first_analysis Last Checkpoint: 19 minutes ago

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Truste

Python 3 (ipykernel)

```
[4]: print("=" * 60)
print("📄 INFORMATION AVAILABLE FOR EACH CURRENCY")
print("=" * 60)

for i, column in enumerate(df.columns, 1):
    print(f"{i:2}>. ♦ {column}")

print("-" * 60)
print(f"📄 Total fields: {len(df.columns)} columns")
print("=" * 60)
```

📄 INFORMATION AVAILABLE FOR EACH CURRENCY

1. ♦ _id
2. ♦ symbol
3. ♦ base_currency
4. ♦ base_currency_desc
5. ♦ base_currency_logoid
6. ♦ circulating_supply
7. ♦ crypto_blockchain_ecosystems_tr
8. ♦ crypto_common_categories_tr
9. ♦ crypto_consensus_algorithms_tr
10. ♦ fundamental_currency_code
11. ♦ max_supply
12. ♦ total_supply
13. ♦ tvl
14. ♦ business_description
15. ♦ community
16. ♦ contracts
17. ♦ doc
18. ♦ explorer
19. ♦ has_bonds
20. ♦ has_contracts
21. ♦ instrument_name
22. ♦ is_blacklisted_in_scanner
23. ♦ medium_logo_url
24. ♦ sources
25. ♦ ticker_title
26. ♦ website
27. ♦ community_x_username
28. ♦ community_x_kol_count
29. ♦ base_currency_id
30. ♦ slug
31. ♦ agentId
32. ♦ social
33. ♦ brief_info
34. ♦ depegging_history
35. ♦ mechanism



```

33. * brief_info
34. * depegging_history
35. * mechanism
36. * updatedAt
37. * lastCrawled
38. * countVideos
39. * linkedin
40. * logo
41. * 24h_vol_cmc

```

```
-----
Total fields: 41 columns
=====
```

```

[5]: print("=" * 60)
print(" WEBSITE AVAILABILITY ANALYSIS")
print("=" * 60)

has_website = df['website'].notna().sum()
total = len(df)
percentage = (has_website / total * 100) if total > 0 else 0

print(f" Total currencies      : {total}")
print(f" Currencies with website : {has_website}")
print(f" Coverage                  : {percentage:.1f}%")

print("-" * 60)
print(" ✨ EXAMPLES OF CURRENCIES WITH WEBSITES")
print("-" * 60)

examples = (
    df[df['website'].notna()][['symbol', 'base_currency_desc', 'website']]
    .head(5)
)

print(examples.to_string(index=False))

print("=" * 60)

```

```
=====
 WEBSITE AVAILABILITY ANALYSIS
=====
```

```

Total currencies      : 78
Currencies with website : 75
Coverage              : 96.2%

```

```
-----
 ✨ EXAMPLES OF CURRENCIES WITH WEBSITES
-----
```

```

symbol base_currency_desc website
DJEDUSD      Died [https://died.xvz/. https://coti.io]

```

```
print(examples.to_string(index=False))

print("=" * 60)

=====
WEBSITE AVAILABILITY ANALYSIS
=====
Total currencies      : 78
Currencies with website : 75
Coverage              : 96.2%
=====

EXAMPLES OF CURRENCIES WITH WEBSITES
=====
symbol base_currency_desc website
DJEUSD      Djed [https://djed.xyz/, https://coti.io]
USDUSD      Tether USDT [https://tether.to]
USDCUSD      USDC [https://www.circle.com/en/usdc]
USDS2USD     USDS [https://sky.money/]
USDEEUSD     Ethena USDe [https://www.ethena.fi/]
=====
```

```
[6]: import matplotlib.pyplot as plt

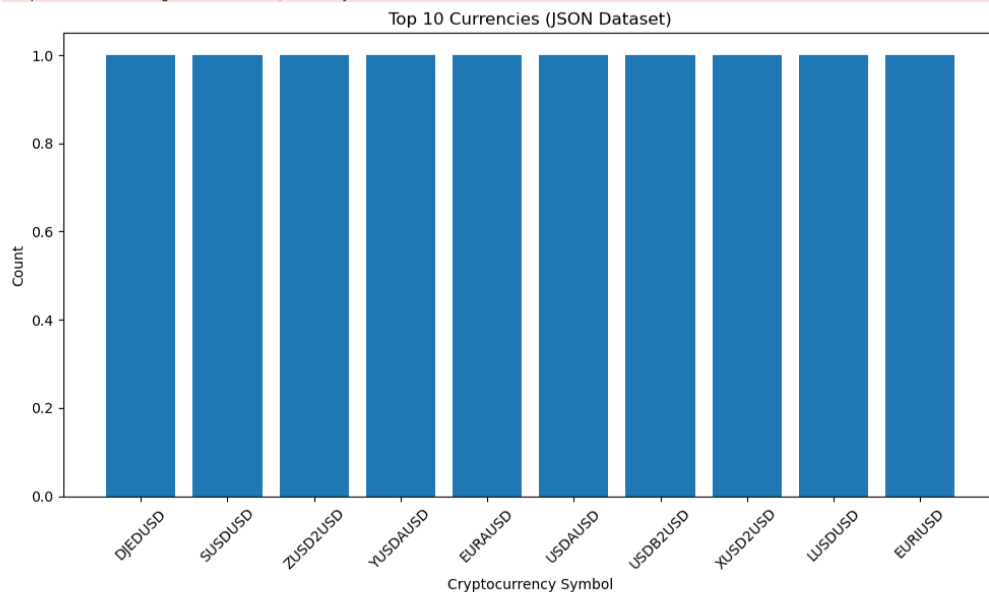
top_currencies = df['symbol'].value_counts().head(10)

plt.figure(figsize=(10, 6))
plt.bar(top_currencies.index, top_currencies.values)
plt.xlabel('Cryptocurrency Symbol')
plt.ylabel('Count')
plt.title('Top 10 Currencies (JSON Dataset)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

print("🎉 Chart created successfully!")
```

Matplotlib is building the font cache; this may take a moment.

Matplotlib is building the font cache; this may take a moment.



🎉 Chart created successfully!

```
[7]: print("📊 QUICK STATS")
print("=" * 40)
print(f"Total currencies: {len(df)}")
print(f"Total columns: {len(df.columns)}")
print(f"With website: {df['website'].notna().sum()}")
print(f"With description: {df['base_currency_desc'].notna().sum()}")
print("=" * 40)

print("🏁 Task 1 completed using JSON!")
```

Jupyter 01_my_first_analysis Last Checkpoint: 20 minutes ago

File Edit View Run Kernel Settings Help Trusted

JupyterLab Python 3 (ipykernel)

```
print("🎉 Task 1 completed using JSON!")
```

QUICK STATS

=====

Total currencies: 78
Total columns: 41
With website: 75
With description: 78
=====

Collapse Output leted using JSON!

```
[8]: df_overview = pd.read_json('v2_token_overview.json')

print(f"📄 Overview records: {len(df_overview)}")
if 'market' in df_overview.columns:
    prices = pd.json_normalize(df_overview['market'])['close']
    print(f"Price (close) - min: {prices.min():.6f}, max: {prices.max():.2f}")
df_overview.head(3)
```

📄 Overview records: 78
Price (close) - min: 0.000160, max: 145.77

[8]:

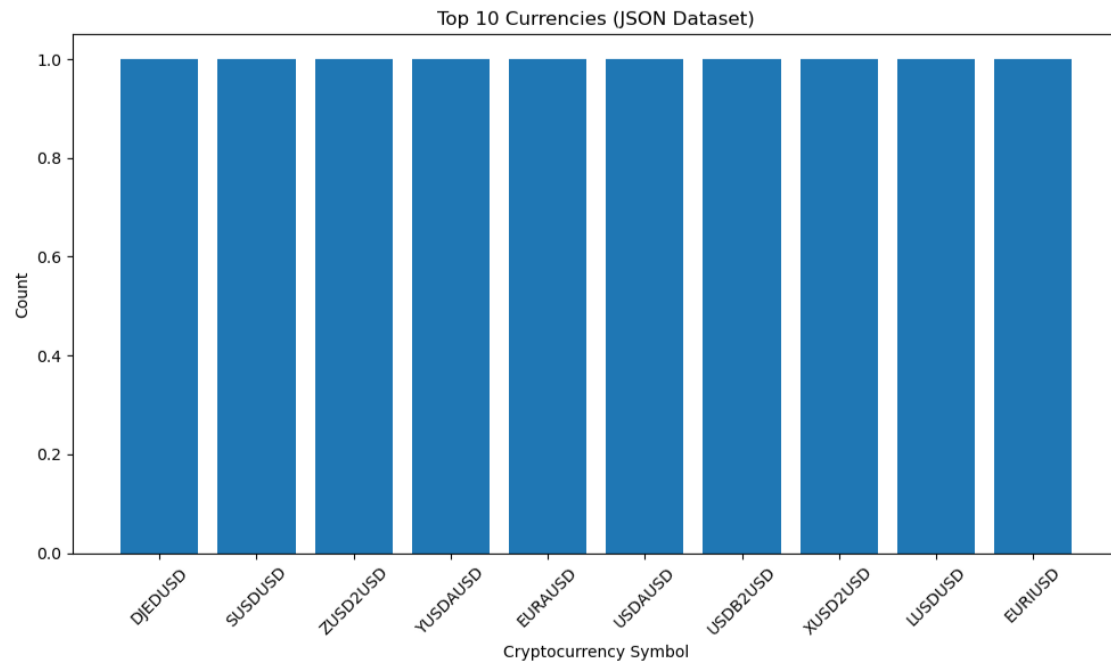
	_id	symbol	holders	info	market	sentiment	transactions
0	67b7640c00bd4e4006b049dd	DJEDUSD	{'active_addresses_ratio': ", 'total_addresse...	{'base_currency': 'DJED', 'base_currency_desc'...	{'change_from_open': 1.259382902377829, 'chang...	{'contributorsactive': 2, 'postsactive': 2, 'a...	{'average_transaction_usd': ", 'fundamental_c...
1	67b7640600bd4e4006b02cce	USDTUSD	{'active_addresses_ratio': ", 'total_addresse...	{'base_currency': 'USDT', 'base_currency_desc'...	{'change_from_open': -0.009998000399918001, 'c...	{'contributorsactive': 357, 'postsactive': 622...	{'average_transaction_usd': 10468.470980220733...
2	67b7640600bd4e4006b02cde	USDCUSD	{'active_addresses_ratio': ", 'total_addresse...	{'base_currency': 'USDC', 'base_currency_desc'...	{'change_from_open': 0.005000850144524001, 'ch...	{'contributorsactive': 1584, 'postsactive': 21...	{'average_transaction_usd': 9558.712262579553,...

3 rows × 26 columns

[]:

2. Your bar chart (the pretty one!)

Matplotlib is building the font cache; this may take a moment.



3. The final statistics output

📊 QUICK STATS

```
=====
Total currencies: 78
Total columns: 41
With website: 75
With description: 78
```